



University of Sri Jayewardenepura

Faculty of Applied Sciences

Department of Mathematics

BSc Degree Second Year End of First Semester Course Unit Examination March, 2021

MAT 211 1.0 Linear Algebra II

Time allowed: 45 minutes

Instructions:

- Answer all questions.
 - Please follow the guidelines given in the **LMS** page.
 - Start each question on a fresh sheet of paper. Write only on one side.
 - This paper carries a total of 60 marks.
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01. Let $A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

- (i) (a) Write down the eigenvalues of A , by giving reasons. [6]
- (b) **Deduce** that the matrix A is not invertible. [3]
- (ii) Use (i)(a) to list the eigenvalues of the matrices $5A$ and $A - kI$. Here k is a fixed constant and I is the identity matrix of order 3. [6]
- (iii) Obtain the eigenvalues of the matrix A^{2021} . [4]

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02. (i) Write the quadratic form $f(x, y) = 6x^2 + 5xy - 6y^2$ in the matrix form $f(x, y) = X^T AX$, where A is a symmetric matrix. [5]

(ii) Give reasons why the matrix A is orthogonally diagonalizable. [6]

(iii) **Hence** identify the conic section of $6x^2 + 5xy - 6y^2 - 7 = 0$. [15]

03. Construct an orthonormal basis for the subspace $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \middle| 2x - y - z = 0 \right\}$ of $\mathbb{R}^3$. [15]

